DFT and circular convolution

Practice questions · Day 09 · Sun 1 Nov · pack D5

5. Practice

From your course files

T5 Q1 (tutorial). Determine the circular convolution of the sequences

$$x_1(n) = \{\underset{\uparrow}{1}, 2, 3, 1\} \qquad x_2(n) = \{\underset{\uparrow}{4}, 3, 2, 2\}$$

using the time-domain circular convolution formula.

T5 Q2 (tutorial). Repeat the above problem using the 4-point DFT and IDFT.

T5 Q3 (tutorial). Compute the 8-point DFT of the sequence

$$x(n) = \begin{cases} 1, & 0 \le n \le 3 \\ 0, & 4 \le n \le 7 \end{cases}$$

Using the DFT obtained above, compute the DFT of the following sequences

$$x_1(n) = \begin{cases} 1, & n = 0 \\ 0, & 1 \le n \le 4 \\ 1, & 5 \le n \le 7 \end{cases} \qquad x_2(n) = \begin{cases} 0, & 0 \le n \le 1 \\ 1, & 2 \le n \le 5 \\ 0, & 6 \le n \le 7 \end{cases}$$

T5 Q6 (tutorial).

(a) Determine the discrete-time Fourier transform $X(\omega)$ of the signal $x(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1, 0\}$.

(b) Compute the 6-point DFT $V(k)$ of the signal $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.

(c) What is the relationship between $X(\omega)$ and $V(k)$? Explain.

T5 Q4 (tutorial). Consider a causal LTI system with transfer function

$$H(z) = \frac{1}{1 - 0.5z^{-1}}.$$

The output signal of the system, $y(n)$, is known for $0 \leq n \leq 63$. Assuming that $H(z)$ is also known, develop a method to recover the input signal $x(n)$ for $0 \leq n \leq 63$ using the 64-point DFT and IDFT.

New questions

Q1 (easy) [3 marks]. Compute the 4-point DFT of $x(n) = \{1, 0, 2, 1\}$ using the $\mathbf{W}_4$ matrix. Verify that $X(3) = X^*(1)$.

Q2 [4 marks]. $x(n) = \{1, 2, 3, 4, 5, 6\}$, $n = 0, \ldots, 5$.

a. Its DTFT is sampled at $\omega_k = 2\pi k/4$, $k = 0, \ldots, 3$. Find the sequence $\tilde{x}(n)$ returned by the 4-point IDFT of these samples. [2]

b. Find $\tilde{X}(0)$ and confirm it equals $X(\omega)$ at $\omega = 0$. [1]

c. What is the smallest N for which the IDFT returns $x(n)$ exactly? [1]

Q3 [5 marks]. $x_1(n) = \{1, 2, 0, 1\}$, $x_2(n) = \{1, 1, 1\}$.

a. Compute $x_1 \circledast_4 x_2$ by the matrix method. [3]
b. Compute the linear convolution and show how (a) follows from it. [2]

Q4 [6 marks, chained]. An FIR filter $h(n) = \{1, -1\}$ is driven by $x(n) = \{1, 2, 1\}$.

a. What is the minimum DFT length for the DFT method to give the linear convolution? [1]
b. Compute the 4-point DFTs $X(k)$ and $H(k)$. [2]
c. Form $Y(k) = X(k)H(k)$ and find $y(n)$ by IDFT. [2]
d. Explain why $Y(k) = Y(\omega)|_{\omega = \pi k/2}$ here, and what $y(n)$ would be with $N = 3$. [1]

Q5 [6 marks]. $x(n) = \{1, 2, 3, 4\}$ has 4-point DFT $X(k) = \{10, -2 + 2j, -2, -2 - 2j\}$. Without computing any new DFT sum, find the 4-point DFT of:

a. $x_a(n) = x((n-1))_4$ [2]
b. $x_b(n) = x((-n))_4$ [2]
c. $x_c(n) = (-1)^n x(n)$ [1]
d. Use Parseval to find $\sum_n x^2(n)$ from $X(k)$. [1]

Q6 (hard) [6 marks]. A sequence $x(n)$ of length 3 was passed through $h(n) = \{2, 1\}$, giving $y(n) = \{2, 5, 6, 2\}$.

a. Recover $x(n)$ using 4-point DFTs. [4]
b. Explain why the same method would fail if $h(n) = \{1, 1\}$ and $N = 4$. [2]

6. Quick recall check

1. What is $((-3))_5$?
2. Write the second row of $\mathbf{W}_4$.
3. Minimum DFT length to linearly convolve sequences of length 10 and 5?
4. Real $x(n)$, $N = 8$, $X(3) = 1 + 2j$. What is $X(5)$?
5. Direct DFT vs radix-2 FFT complex multiplications for $N = 256$?